

SNx4HCT125 Quadruple Bus Buffer Gates With 3-State Outputs

1 Features

- Operating voltage range of 4.5 V to 5.5 V
- High-current can drive up to 15 LSTTL loads
- Low power consumption, 80- μ A max I_{CC}
- Typical $t_{pd} = 12$ ns
- ± 6 -mA output drive at 5 V
- Low input current of 1 μ A max
- Inputs are TTL-voltage compatible
- High-current 3-state outputs drive bus lines or buffer memory address registers

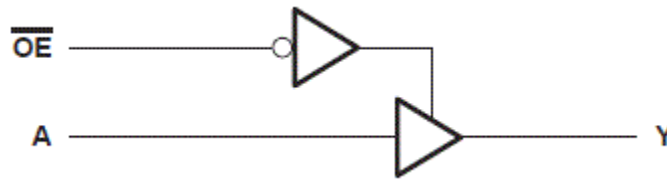
2 Description

The SNx4HCT125 contains four independent buffers with TTL-compatible inputs and 3-state outputs. Each gate performs the Boolean function $Y = A$ in positive logic.

Device Information

PART NUMBER	PACKAGE ⁽¹⁾	BODY SIZE (NOM)
SN74HCT125D	SOIC (14)	8.65 mm $\times$ 3.90 mm
SN74HCT125N	PDIP (14)	19.31 mm $\times$ 6.35 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.



Functional Block Diagram



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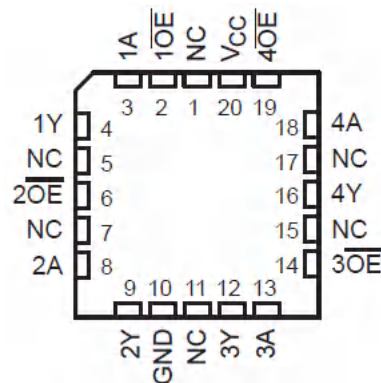
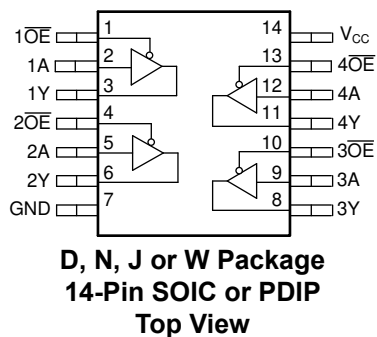
3 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision F (February 2022) to Revision G (October 2022)	Page
• Increased R _{θJA} for packages: D (86 to 138.7); N (80 to 75.3).....	4

Changes from Revision E (August 2003) to Revision F (February 2022)	Page
• Updated the numbering, formatting, tables, figures, and cross-references throughout the document to reflect modern data sheet standards.....	1

4 Pin Configuration and Functions



NC – No internal connection

5 Specifications

5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)⁽¹⁾

		MIN	MAX	UNIT
V _{CC}	Supply voltage range	-0.5	7	V
I _{IK}	Input clamp current ⁽²⁾	(V _I < 0 or V _I > V _{CC})	±20	mA
I _{OK}	Output clamp current ⁽²⁾	(V _O < 0 or V _O > V _{CC})	±20	mA
I _O	Continuous output current	(V _O = 0 to V _{CC})	±35	mA
V _{CC} or GND	Continuous current through		±70	mA
T _J	Junction temperature		150	°C
T _{stg}	Storage temperature	-65	150	°C

- (1) Stresses beyond those listed under *absolute maximum ratings* may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these or any other conditions beyond those indicated under *recommended operating conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

5.2 Recommended Operating Conditions⁽¹⁾

			SN54HCT125 ⁽²⁾			SN74HCT125			UNIT
			MIN	NOM	MAX	MIN	NOM	MAX	
V _{CC}	Supply voltage		4.5	5	5.5	4.5	5	5.5	V
V _{IH}	High-level input voltage	V _{CC} = 4.5 V to 5.5 V	2			2			V
V _{IL}	Low-level input voltage	V _{CC} = 4.5 V to 5.5 V			0.8			0.8	V
V _I	Input voltage		0		V _{CC}	0		V _{CC}	V
V _O	Output voltage		0		V _{CC}	0		V _{CC}	V
t _i	Input transition rise/fall time				500			500	ns
T _A	Operating free-air temperature		-55		125	-40		85	°C

- (1) All unused inputs of the device must be held at V_{CC} or GND to ensure proper device operation. Refer to the TI application report Implications of Slow or Floating SMOS Inputs, literature number [SCBA004](#).
- (2) SN54HCT125 is in product preview.

5.3 Thermal Information

THERMAL METRIC		D (SOIC)	N (PDIP)	UNIT
		14 PINS	14 PINS	
R _{θJA}	Junction-to-ambient thermal resistance ⁽¹⁾	138.7	75.3	°C/W
R _{θJC}	Junction-to-case (top) thermal resistance	93.8	68.6	°C/W
R _{θJB}	Junction-to-board thermal resistance	94.7	55.1	°C/W
Ψ _{JT}	Junction-to-top characterization parameter	49.1	41.1	°C/W
Ψ _{JB}	Junction-to-board characterization parameter	94.3	54.9	°C/W
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	N/A	N/A	°C/W

- (1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC package thermal metrics](#) application report.

5.4 Electrical Characteristics

PARAMETER		TEST CONDITIONS ⁽¹⁾	V _{CC} (V)	T _A = 25°C			SN54HCT125 ⁽³⁾		SN74HCT125		UNIT
				MIN	TYP	MAX	MIN	MAX	MIN	MAX	
V _{OH}	High-level output voltage	I _{OH} = –20 µA	4.5	4.4	4.499		4.4		4.4		V
		I _{OH} = –6 mA		3.98	4.3		3.7		3.84		
V _{OL}	Low-level output voltage	I _{OL} = 20 µA	5.5		0.001	0.1		0.1		0.1	V
		I _{OL} = 6 mA			0.17	0.26		0.4		0.33	
I _I	Input hold current	V _I = V _{CC} or 0	5.5		±0.1	±100		±1000		±1000	nA
I _{OZ}	Off-state output current	V _O = V _{CC} or 0	5.5		±0.01	±0.5		±10		±5	µA
I _{CC}	Supply current	V _I = V _{CC} or 0. I _O = 0	5.5			8		160		80	µA
ΔI _{CC} ⁽²⁾	Supply-current change	One input at 0.5 V or 2.4 V, Other inputs at 0 or V _{CC}	5.5		1.4	2.4		3		2.9	mA
C _i	Input capacitance		4.5 to 5.5		3	10		10 ⁽⁴⁾		10	pF

(1) V_I = V_{IH} or V_{IL}, unless otherwise noted.

(2) This is the increase in supply current for each input that is at one of the specified TTL voltage levels, rather than 0 V or V_{CC}.

(3) SN54HCT125 is in product preview.

(4) On products compliant to MIL-PRF-38535, this parameter is not production tested.

5.5 Switching Characteristics

C_L = 50 pF. See [Figure 6](#)

PARAMETER		FROM (INPUT)	TO (OUTPUT)	V _{CC} (V)	T _A = 25°C			SN54HCT125 ⁽¹⁾		SN74HCT125		
					MIN	TYP	MAX	MIN	MAX	MIN	MAX	
t _{pd}	Propagation delay	A	Y	4.5		11	20		39		25	ns
				5.5		10	18		35		22	
t _{en}	Enable time	OE	Y	4.5		18	28		42		35	ns
				5.5		15	25		38		31	
t _{dis}	Diable time	OE	Y	4.5		15	26		39		33	ns
				5.5		13	23		35		30	
t _t	Transition time		Any	4.5		8	15		22		19	ns
				5.5		7	14		21		17	

(1) SN54HCT125 is in product preview.

5.5 Switching Characteristics

C_L = 150 pF. See [Figure 6](#)

PARAMETER		FROM (INPUT)	TO (OUTPUT)	V _{CC} (V)	T _A = 25°C			SN54HCT125 ⁽¹⁾		SN74HCT125		
					MIN	TYP	MAX	MIN	MAX	MIN	MAX	
t _{pd}	Propagation delay	A	Y	4.5		19	36		58		46	ns
				5.5		16	32		48		42	
t _{en}	Enable time	OE	Y	4.5		25	40		60		50	ns
				5.5		21	35		53		43	
t _t	Transition time		Any	4.5		17	42		63		53	ns
				5.5		14	38		57		48	

(1) SN54HCT125 is in product preview.

5.6 Operating Characteristics

 $T_A = 25^\circ\text{C}$

		Test Conditions	TYP	UNIT
C_{pd}	Power dissipation capacitance	No load	35	pF

6 Parameter Measurement Information

t_{pd} is the maximum between t_{PLH} and t_{PHL}

t_t is the maximum between t_{TLH} and t_{THL}

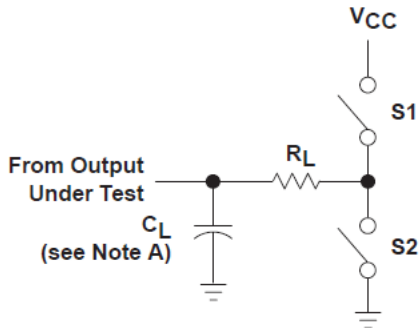


Figure 6-1. Load Circuit

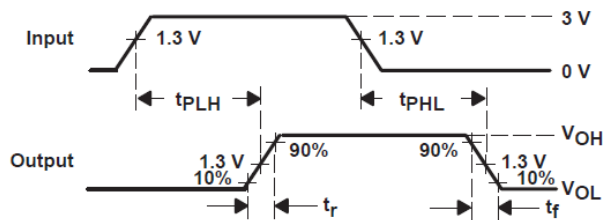


Figure 6-3. Voltage Waveforms Propagation Delay Times

PARAMETER		R_L	C_L	S1	S2
t_{en}	t_{PZH}	1 k Ω	50 pF or 150 pF	Open	Closed
	t_{PZL}			Closed	Open
t_{dis}	t_{PHZ}	1 k Ω	50 pF	Open	Closed
	t_{PLZ}			Closed	Open
t_{pd} or t_t		—	50 pF or 150 pF	Open	Open

Figure 6-2.

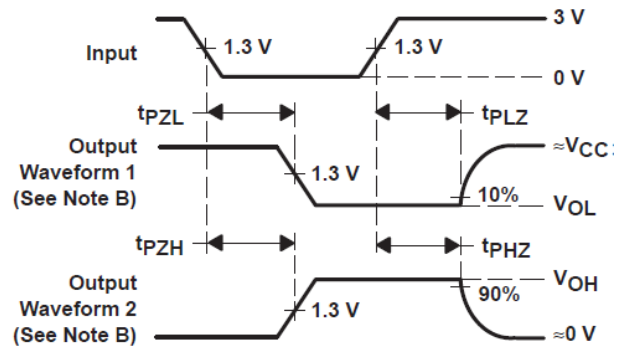


Figure 6-4. Voltage Waveforms Enable and Disable Times For 3-state Outputs

A. C_L includes probe and jig capacitance.

B. Waveform 1 is for an output with internal conditions such that the output is low except when disabled by the output control.

Waveform 2 is for an output with internal conditions such that the output is high except when disabled by the output control.

C. Phase relationships between waveforms were chosen arbitrarily. All input pulses are supplied by generators having the following characteristics: $PRR \leq 1$ MHz, $Z_O = 50 \Omega$, $t_r = 6$ ns, $t_f = 6$ ns.

D. The outputs are measured one at a time with one input transition per measurement.

E. t_{PLZ} and t_{PHZ} are the same as t_{dis} .

F. t_{PZL} and t_{PZH} are the same as t_{en} .

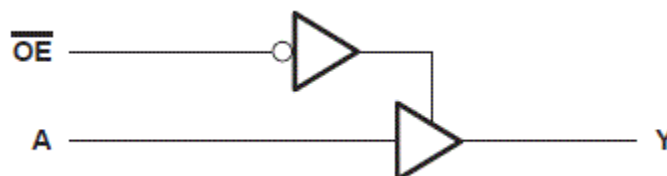
7 Detailed Description

7.1 Overview

These bus buffer gates feature independent line drivers with 3-state outputs. Each output is disabled when the associated output-enable ($\overline{OE}$) input is high.

To ensure the high-impedance state during power up or power down, $\overline{OE}$ should be tied to V_{CC} through a pullup resistor; the minimum value of the resistor is determined by the current-sinking capability of the driver.

7.2 Functional Block Diagram



7.3 Device Functional Modes

**Table 7-1. Function Table
(each gate)**

INPUTS		OUTPUT
$\overline{OE}$	A	Y
L	H	H
L	L	L
H	X	Z

8 Power Supply Recommendations

The power supply can be any voltage between the minimum and maximum supply voltage rating located in the *Recommended Operating Conditions*. Each V_{CC} terminal should have a good bypass capacitor to prevent power disturbance. A 0.1- μ F capacitor is recommended for this device. It is acceptable to parallel multiple bypass caps to reject different frequencies of noise. The 0.1- μ F and 1- μ F capacitors are commonly used in parallel. The bypass capacitor should be installed as close to the power terminal as possible for best results.

9 Layout

9.1 Layout Guidelines

When using multiple-input and multiple-channel logic devices inputs must not ever be left floating. In many cases, functions or parts of functions of digital logic devices are unused; for example, when only two inputs of a triple-input AND gate are used or only 3 of the 4 buffer gates are used. Such unused input pins must not be left unconnected because the undefined voltages at the outside connections result in undefined operational states. All unused inputs of digital logic devices must be connected to a logic high or logic low voltage, as defined by the input voltage specifications, to prevent them from floating. The logic level that must be applied to any particular unused input depends on the function of the device. Generally, the inputs are tied to GND or V_{CC} , whichever makes more sense for the logic function or is more convenient.

10 Device and Documentation Support

TI offers an extensive line of development tools. Tools and software to evaluate the performance of the device, generate code, and develop solutions are listed below.

10.1 Documentation Support

10.1.1 Related Documentation

10.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Subscribe to updates* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

10.3 Support Resources

TI E2E™ [support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

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10.4 Trademarks

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10.5 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

10.6 Glossary

[TI Glossary](#) This glossary lists and explains terms, acronyms, and definitions.

11 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
SN74HCT125D	Obsolete	Production	SOIC (D) 14	-	-	Call TI	Call TI	-40 to 85	HCT125
SN74HCT125DR	Active	Production	SOIC (D) 14	2500 LARGE T&R	Yes	NIPDAU SN	Level-1-260C-UNLIM	-40 to 85	HCT125
SN74HCT125DR.A	Active	Production	SOIC (D) 14	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	HCT125
SN74HCT125DRE4	Active	Production	SOIC (D) 14	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	HCT125
SN74HCT125DRE4.A	Active	Production	SOIC (D) 14	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	HCT125
SN74HCT125DT	Obsolete	Production	SOIC (D) 14	-	-	Call TI	Call TI	-40 to 85	HCT125
SN74HCT125N	Active	Production	PDIP (N) 14	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 85	SN74HCT125N
SN74HCT125N.A	Active	Production	PDIP (N) 14	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 85	SN74HCT125N
SN74HCT125NE4	Active	Production	PDIP (N) 14	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	-40 to 85	SN74HCT125N

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

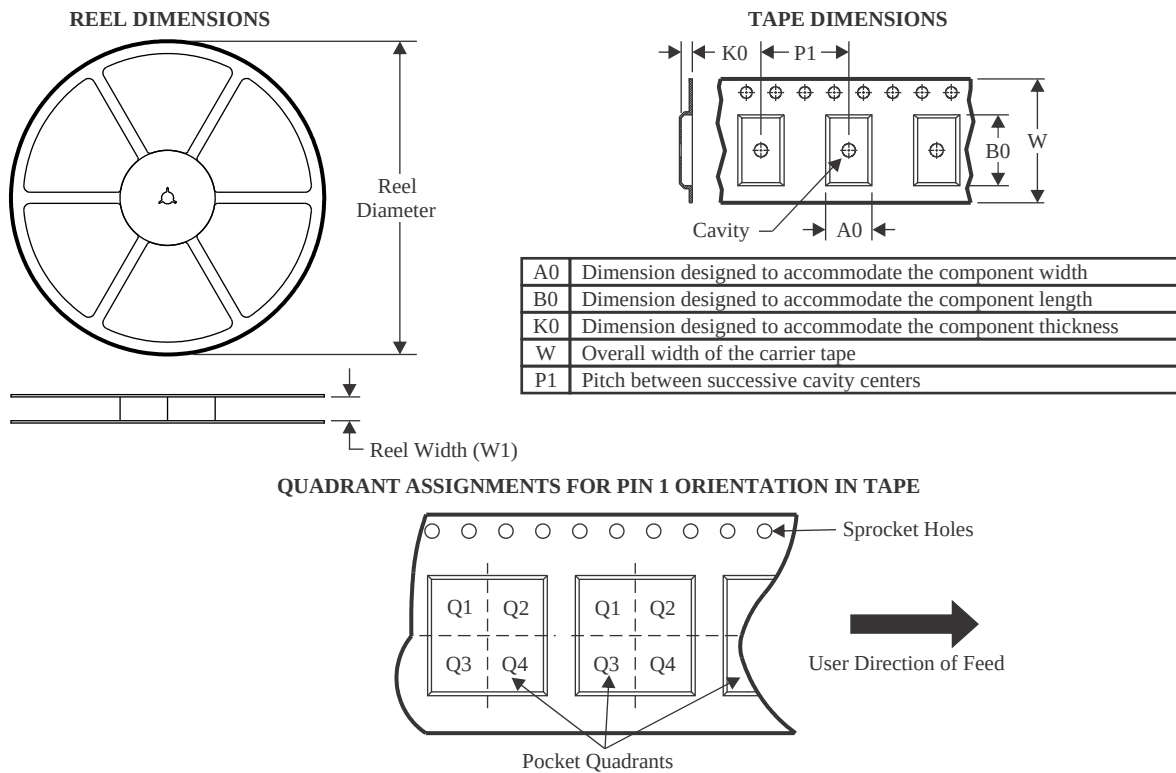
Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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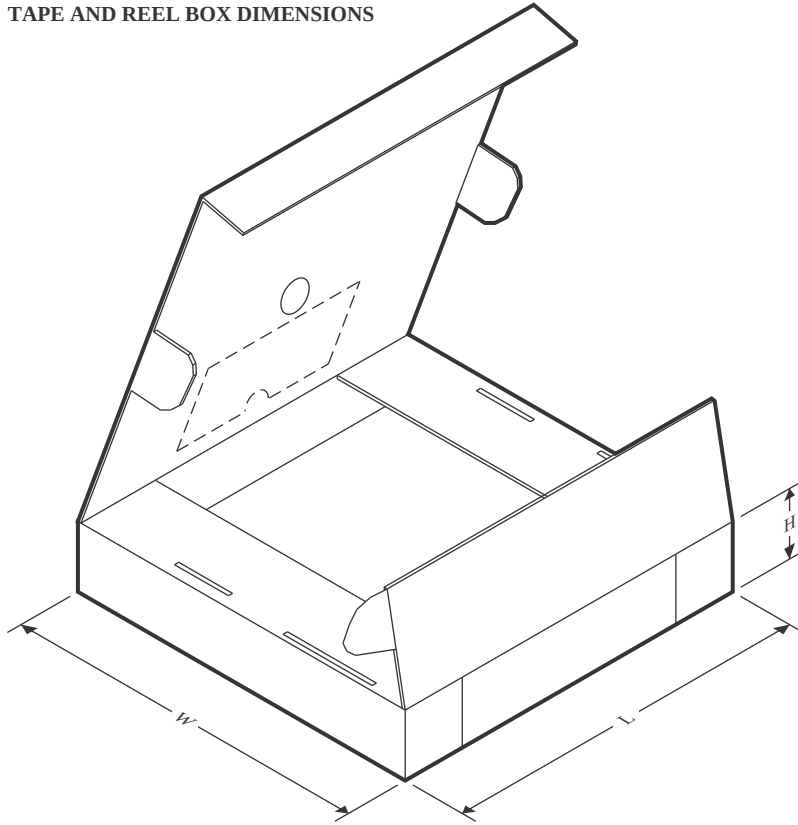
TAPE AND REEL INFORMATION



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
SN74HCT125DR	SOIC	D	14	2500	330.0	16.4	6.5	9.0	2.1	8.0	16.0	Q1
SN74HCT125DRE4	SOIC	D	14	2500	330.0	16.4	6.5	9.0	2.1	8.0	16.0	Q1

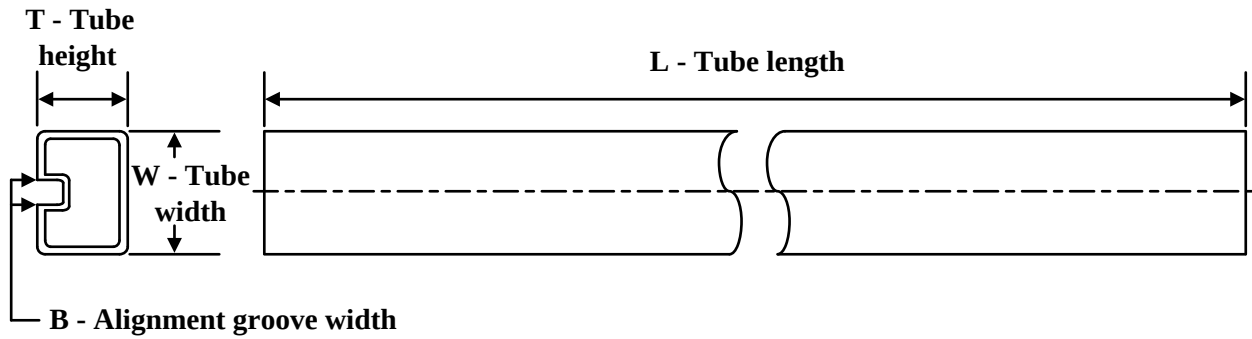
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
SN74HCT125DR	SOIC	D	14	2500	353.0	353.0	32.0
SN74HCT125DRE4	SOIC	D	14	2500	353.0	353.0	32.0

TUBE



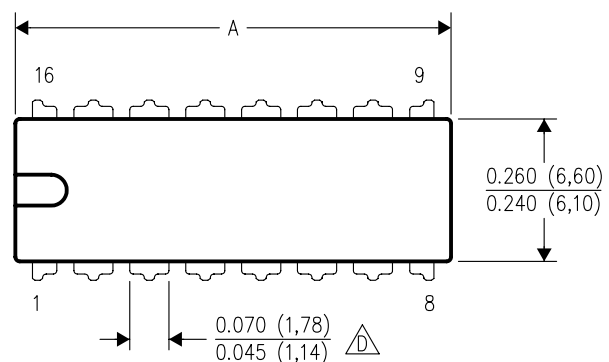
*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
SN74HCT125N	N	PDIP	14	25	506	13.97	11230	4.32
SN74HCT125N	N	PDIP	14	25	506	13.97	11230	4.32
SN74HCT125N.A	N	PDIP	14	25	506	13.97	11230	4.32
SN74HCT125N.A	N	PDIP	14	25	506	13.97	11230	4.32
SN74HCT125NE4	N	PDIP	14	25	506	13.97	11230	4.32
SN74HCT125NE4	N	PDIP	14	25	506	13.97	11230	4.32

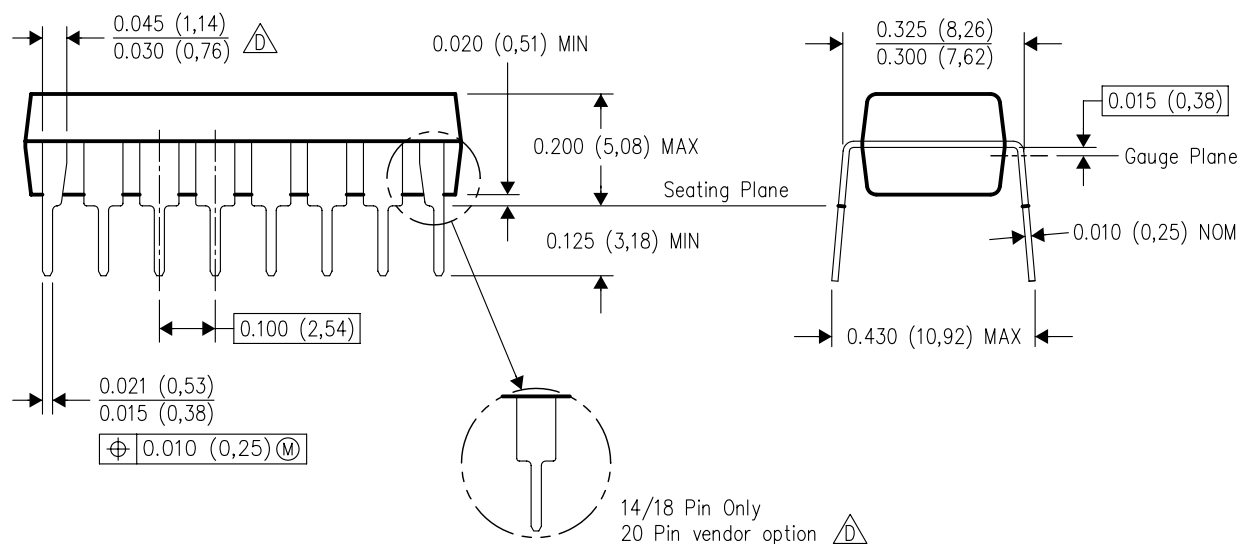
N (R-PDIP-T**)

16 PINS SHOWN

PLASTIC DUAL-IN-LINE PACKAGE

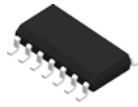


PINS **	14	16	18	20
DIM				
A MAX	0.775 (19,69)	0.775 (19,69)	0.920 (23,37)	1.060 (26,92)
A MIN	0.745 (18,92)	0.745 (18,92)	0.850 (21,59)	0.940 (23,88)
MS-001 VARIATION	AA	BB	AC	AD

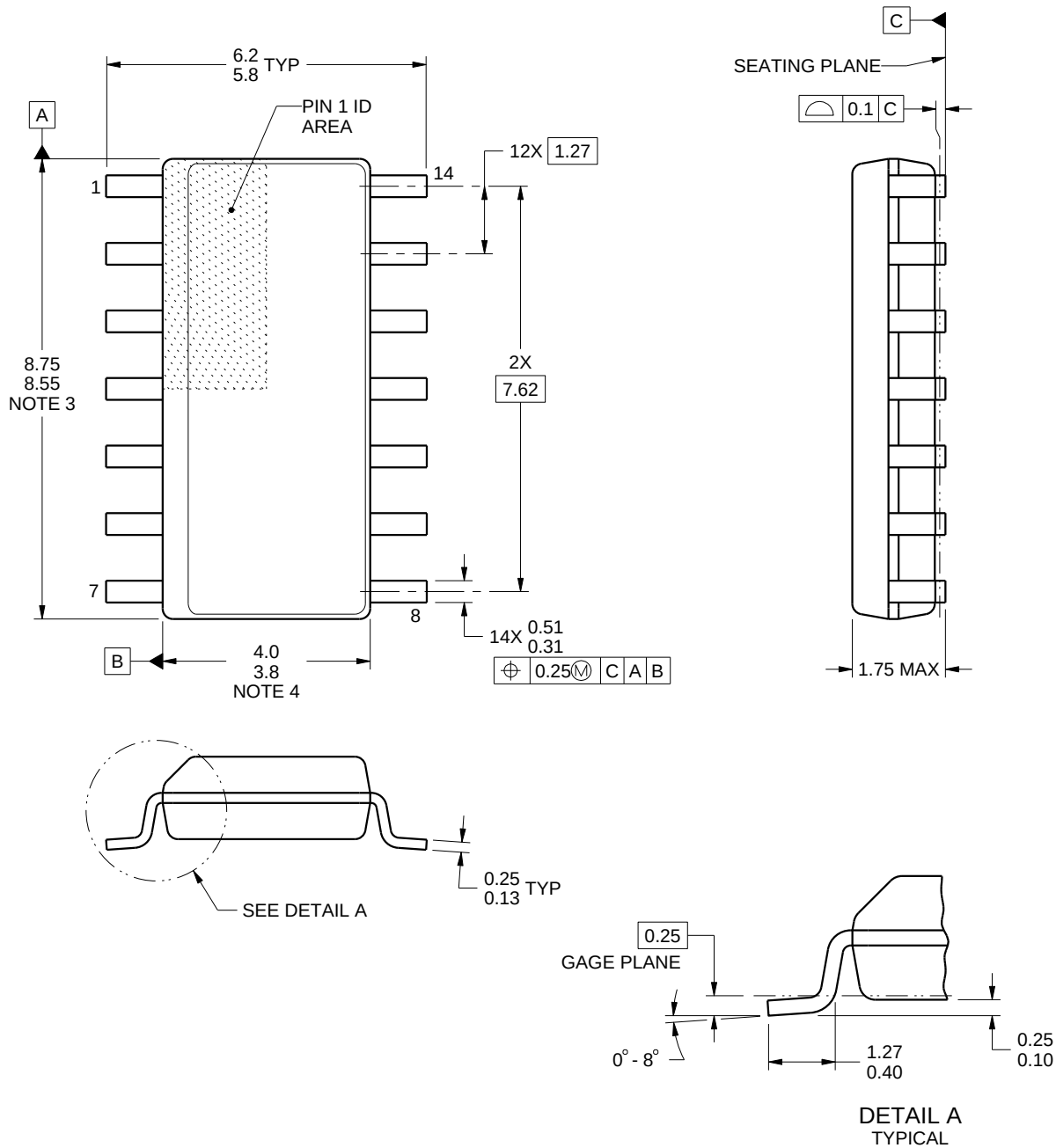


4040049/E 12/2002

- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 - Falls within JEDEC MS-001, except 18 and 20 pin minimum body length (Dim A).
 - The 20 pin end lead shoulder width is a vendor option, either half or full width.

D0014A**PACKAGE OUTLINE****SOIC - 1.75 mm max height**

SMALL OUTLINE INTEGRATED CIRCUIT



4220718/A 09/2016

NOTES:

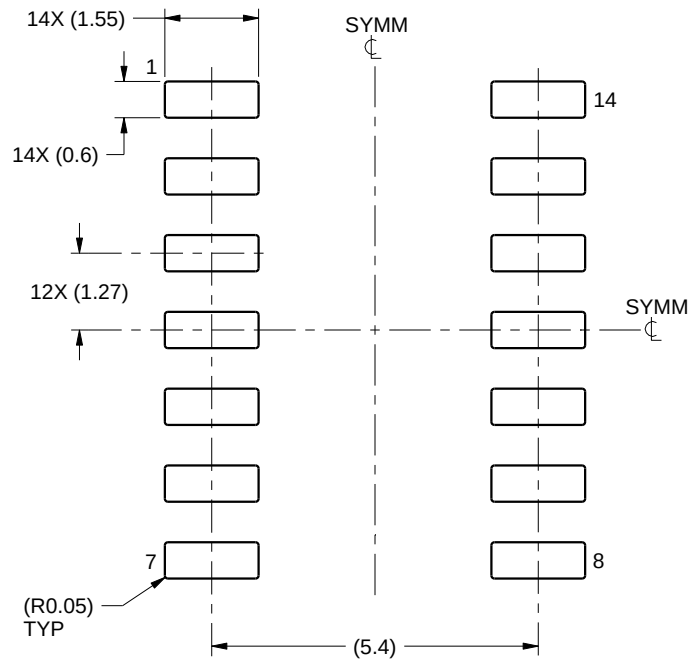
1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.43 mm, per side.
5. Reference JEDEC registration MS-012, variation AB.

EXAMPLE BOARD LAYOUT

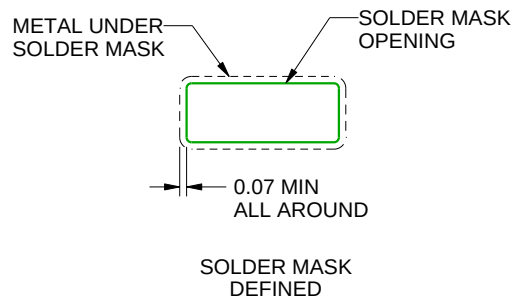
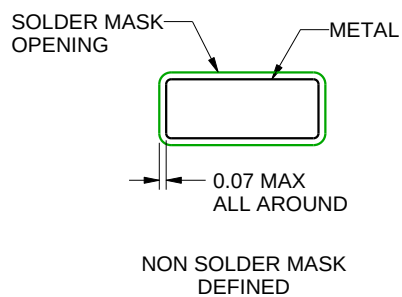
D0014A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



LAND PATTERN EXAMPLE
SCALE:8X



SOLDER MASK DETAILS

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NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

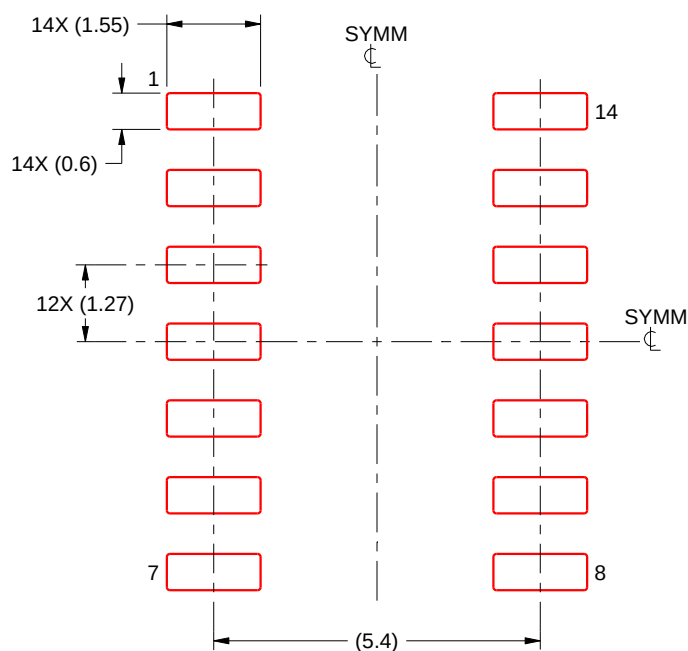
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

D0014A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:8X

4220718/A 09/2016

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

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